

CH1101: Quantum Chemistry & Spectroscopy Lec05

Time-independent

Schrödinger Eq. (TISE)

$$\hat{H}\Psi = E\Psi$$

24/08/2026

$\hat{H}$: Hamiltonian operator (hermitian & linear)

Ψ : wavefunction

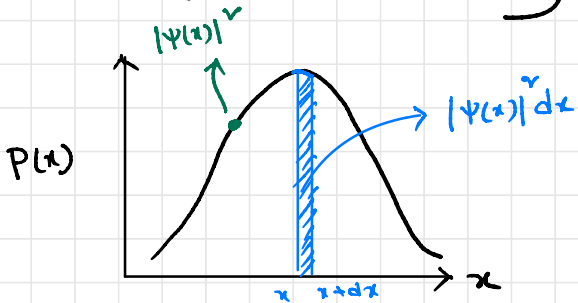
E : Total energy

By expanding $\Rightarrow$

$$\underbrace{\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right]}_{\hat{H}} \Psi(x) = E \Psi(x)$$

Probability density : $P(x) = |\psi(x)|^2 = \psi^*(x) \psi(x)$

Probability of finding
a particle in space
of region x to $x+dx$ } :



$$\int_x^{x+dx} P(x) dx$$

$$= \int_x^{x+dx} |\psi(x)|^2 dx$$

$$= \int_x^{x+dx} \psi^*(x) \psi(x) dx$$

Complex no.

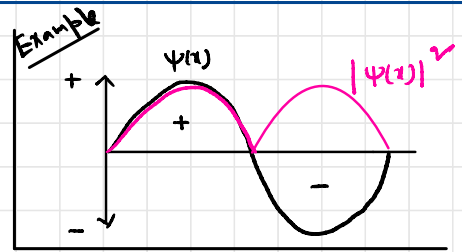
$$z = x + iy$$

$$z^* = x - iy$$

(complex conjugate of z)

$$|z|^2 = z \times z^* = (x + iy)(x - iy)$$

$$= x^2 + y^2$$



For example

$$\psi = e^{-ikx}$$

$$\psi^* = e^{+ikx}$$

Therefore,

$$P(x) = \psi^* \psi$$

$$= e^{ikx} e^{-ikx}$$

Model problems can be solved by using

TISE

Types of motion

1. Free-particle

2. Particle in one-dimensional box

3. Particle in a ring

Translational

Rotational

Vibrational

FREE PARTICLE

$$\hat{H}\psi = E\psi$$

TISE

$$\Rightarrow \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \underbrace{V(x)}_0 \right] \psi(x) = E\psi(x)$$

For $-\infty < x < +\infty$
for free-particle

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = E\psi(x)$$

$$\Rightarrow \frac{d^2 \psi(x)}{dx^2} + \underbrace{\frac{2mE}{\hbar^2}}_{k^2} \psi(x) = 0$$

NOTE

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0$$

$$k^2 = \frac{2mE}{\hbar^2} \Rightarrow \boxed{\overset{\text{Energy}}{E} = \frac{\hbar^2 k^2}{2m}}$$

For k of real values
including ZERO!

As k can have any real values, Energy
is nothing but Continuous!

Free particle is not a quantum particle!